1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In TMPRSS2/ERG fusion–positive prostate tumors, constitutive PI3K signaling increases tumor cell PD-L1 (CD274) expression, and TLS-like lymphoid aggregates form but fail to drive durable anti-tumor immunity due to PD-L1–mediated T cell suppression within the tumor microenvironment. Consequently, TMPRSS2/ERG+ tumors with high CD274 display reduced responsiveness to PD-1/PD-L1 blockade relative to TMPRSS2/ERG– or CD274-low tumors, even when TLS-like structures are present. Disrupting the TMPRSS2/ERG axis or inhibiting PI3K signaling decreases CD274 levels, reinvigorates TLS-associated T cell activity, and enhances responsiveness to PD-1/PD-L1 inhibitors. PSMA/FOLH1 expression and PI3K pathway activity act as secondary modifiers that modulate the magnitude but do not define the core anchor relationship between TMPRSS2/ERG and CD274, thereby shaping but not overturning the predicted resistance to immune checkpoint therapy. This yields testable predictions about (a) the PI3K–CD274 link in TMPRSS2/ERG+ cells, (b) the functional state of TLS-associated T cells, and (c) the efficacy of combination therapies (PD-1/PD-L1 blockade plus PI3K inhibition or TMPRSS2/ERG disruption) specifically within TLS-rich tumor regions.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CD274, TMPRSS2/ERG
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD274: CD274 upregulation downstream of TMPRSS2/ERG/PI3K) directly predicts altered sensitivity to PD-1/PD-L1–directed therapies (and may modulate BCG-related immune contexts).

TMPRSS2/ERG: TMPRSS2/ERG/PI3K) directly predicts altered sensitivity to PD-1/PD-L1–directed therapies (and may modulate BCG-related immune contexts).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	FOLH1 (PSMA)	Supporting modifier from knowledge graph

Secondary	PIK3CA (PI3K pathway activation)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: TMPRSS2/ERG fusions are prevalent in prostate cancer and can co-activate PI3K/AKT signaling;

Prediction 2: PD-L1 (CD274) expression can be driven by PI3K/AKT activity, providing a direct mechanism for immune evasion in TMPRSS2/ERG+ tumors; and

Prediction 3: TLS-like structures frequently appear in these tumors but do not guarantee effective anti-tumor immunity, suggesting that intrinsic tumor cell–mediated immune suppression may dominate over TLS architecture. The expanded hypothesis formalizes a testable axis: TMPRSS2/ERG → PI3K → CD274 controls TLS-associated immune efficacy and therapy response.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- FOLH1 (PSMA): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PIK3CA (PI3K pathway activation): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.